

Arjuna JEE 2025

Maths

DPP: 2

Trigonometric Equations

Q1 Find the general solution of the following equation: $\sqrt{3} \sec \theta = 2$

- (A) $\theta = 2n\pi \pm \frac{\pi}{3}$
 (B) $\theta = 2n\pi \pm \frac{\pi}{6}$
 (C) $\theta = 2n\pi \pm \frac{5\pi}{6}$
 (D) $\theta = 2n\pi \pm \frac{7\pi}{6}$

Q2 Find the general solution of the following equation: $\cos 3\theta = -\frac{1}{2}$

- (A) $2n\pi \pm \frac{2\pi}{3}$
 (B) $2n\pi \pm \frac{\pi}{3}$
 (C) $\frac{2n\pi}{3} \pm \frac{2\pi}{9}$
 (D) $\frac{2n\pi}{3} \pm \frac{4\pi}{3}$

Q3 Find the general solution of the following equation: $2 \sin \theta + 1 = 0$

- (A) $n\pi + (-1)^n \frac{\pi}{6}$
 (B) $n\pi + (-1)^{n+1} \frac{\pi}{6}$
 (C) $n\pi + (-1)^n \frac{5\pi}{6}$
 (D) $n\pi + (-1)^n \frac{\pi}{3}$

Q4 If $4 \sin^2 \theta = 1$, then the value of θ are-

- (A) $2n\pi \pm \frac{\pi}{3}, n \in \mathbb{Z}$
 (B) $n\pi \pm \frac{\pi}{3}, n \in \mathbb{Z}$
 (C) $n\pi \pm \frac{\pi}{6}, n \in \mathbb{Z}$
 (D) $2n\pi \pm \frac{\pi}{6}, n \in \mathbb{Z}$

Q5 Find the general solution of the equation $\tan 3\theta = -1$

- (A) $n\pi + \left(-\frac{\pi}{4}\right)$
 (B) $n\pi + \frac{\pi}{4}$
 (C) $\frac{n\pi}{3} - \frac{\pi}{12}$
 (D) $n\pi + \frac{\pi}{12}$

Q6 Find the principle value of θ . for $2 \sin^2 \theta + \sqrt{3} \cos \theta + 1 = 0$

- (A) $\frac{5\pi}{6}$
 (B) $\frac{2\pi}{3}$
 (C) $\pi/3$
 (D) $\pi/6$

Q7 If $\sin^2 \theta - 2 \cos \theta + \frac{1}{4} = 0$, then the general value of θ is

- (A) $n\pi \pm \frac{\pi}{3}$
 (B) $2n\pi \pm \frac{\pi}{3}$
 (C) $2n\pi \pm \frac{\pi}{6}$
 (D) $n\pi \pm \frac{\pi}{6}$

Q8 The general value θ obtained from the equation $\cos 2\theta = \sin \alpha$, is

- (A) $2\theta = \frac{\pi}{2} - \alpha$
 (B) $\theta = 2n\pi \pm \left(\frac{\pi}{2} - \alpha\right)$
 (C) $\theta = \frac{n\pi + (-1)^n \alpha}{2}$
 (D) $\theta = n\pi \pm \left(\frac{\pi}{4} - \frac{\alpha}{2}\right)$

Q9 If $\cos \theta = -\frac{1}{2}$ and $\tan \theta = \sqrt{3}$, then find the most general value of θ satisfying both the equations.

- (A) $2n\pi + \frac{\pi}{3}$
 (B) $2n\pi + \frac{2\pi}{3}$
 (C) $2n\pi + \frac{4\pi}{3}$
 (D) None

Q10 If $\cos \theta + \sec \theta = \frac{5}{2}$, then the general value of θ is

- (A) $n\pi \pm \frac{\pi}{3}$
 (B) $2n\pi \pm \frac{\pi}{6}$



- (C) $n\pi \pm \frac{\pi}{6}$
 (D) $2n\pi \pm \frac{\pi}{3}$

Q11 Find the value of x , of the given following equation $2\cos^2 x + 3\sin x = 0$

- (A) $n\pi + (-1)^n \frac{\pi}{6}$
 (B) $n\pi + (-1)^n \frac{5\pi}{6}$
 (C) $n\pi + (-1)^n \frac{7\pi}{6}$
 (D) $n\pi + (-1)^n \frac{\pi}{3}$

Q12 Find the value of x : $(2\sin x - \cos x)(1 + \cos x) = \sin^2 x$

- (A) $(2n+1)\frac{\pi}{2}, n\pi + (-1)^n \frac{\pi}{6}$
 (B) $(2n+1)\pi, n\pi + (-1)^{n+1} \frac{\pi}{6}$
 (C) $(2n+1)\pi, n\pi + (-1)^n \frac{\pi}{6}$
 (D) $(2n+1)\frac{\pi}{2}, n\pi + (-1)^{n+1} \frac{\pi}{6}$

Q13 The general value of θ in the equation $\cos \theta = \frac{1}{\sqrt{2}}, \tan \theta = -1$ is

- (A) $2n\pi + \frac{\pi}{6}, n \in \mathbb{Z}$
 (B) $2n\pi + \frac{7\pi}{4}, n \in \mathbb{Z}$
 (C) $n\pi + (-1)^n \frac{\pi}{3}, n \in \mathbb{Z}$
 (D) $n\pi + (-1)^n \frac{\pi}{4}, n \in \mathbb{Z}$

Q14 $\sec^2 2x = 1 - \tan 2x$

- (A) $x = \frac{n\pi}{2}$ or $x = \frac{n\pi}{2} + \frac{3\pi}{8}$
 (B) $x = n\pi$ or $x = \frac{n\pi}{2} + \frac{3\pi}{8}$
 (C) $x = \frac{n\pi}{2}$ or $x = \frac{n\pi}{2} + \frac{3\pi}{4}$
 (D) $x = n\pi$ or $x = \frac{n\pi}{2} + \frac{3\pi}{4}$

Q15 If $\sin^2 \theta = 1/4$, then the most general value of θ is

- (A) $2n\pi \pm (-1)^n \frac{\pi}{6}$
 (B) $\frac{n\pi}{2} \pm (-1)^n \frac{\pi}{6}$
 (C) $n\pi \pm \frac{\pi}{6}$
 (D) $2n\pi \pm \frac{\pi}{6}$

Q16 The general solution of $\sec \theta + \tan \theta = \sqrt{3}$ is

- (A) $\theta = 2n\pi + \frac{\pi}{6}, 2n\pi - \frac{\pi}{2}, n \in \mathbb{I}$
 (B) $\theta = 2n\pi + \frac{\pi}{6}, n \in \mathbb{I}$

(C) $2n\pi - \frac{\pi}{2}, n \in \mathbb{I}$

(D) $2n\pi - \frac{\pi}{2}; n \neq 2k+1, n, k \in \mathbb{I}$



Answer Key

Q1 (B)
Q2 (C)
Q3 (B)
Q4 (C)
Q5 (C)
Q6 (A)
Q7 (B)
Q8 (D)

Q9 (C)
Q10 (D)
Q11 (C)
Q12 (C)
Q13 (B)
Q14 (A)
Q15 (C)
Q16 (B)



[Android App](#) | [iOS App](#) | [PW Website](#)